



Language: **English**

Day 1

Tuesday, February 16, 2021

Problem 1. Let ABC be an isosceles triangle with $BC = CA$, and let D be a point inside side AB such that $AD < DB$. Let P and Q be two points inside sides BC and CA , respectively, such that $\angle DPB = \angle DQA = 90^\circ$. Let the perpendicular bisector of PQ meet line segment CQ at E , and let the circumcircle of triangle ABC and the circumcircle of triangle CPQ meet again at point F , different from C .

Suppose that P, E, F are collinear. Prove that $\angle ACB = 90^\circ$.

Problem 2. Let $f : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ be such that

$$f(x + f(y))^2 \geq f(x)(f(x + f(y)) + f(y))$$

for all $x, y \in \mathbb{R}^+$. Show that f is *unbounded*, i.e. for each $M \in \mathbb{R}^+$, there exists $x \in \mathbb{R}^+$ such that $f(x) > M$.

Problem 3. Let $1 \leq n \leq 2021$ be a positive integer. Jack has 2021 coins arranged in a line where each coin has an H on one side and a T on the other. At the beginning, all coins show H except the n^{th} coin. Jack can repeatedly perform the following operation: he chooses a coin showing T , and turns over the coins next to it to the left and to the right (if any).

Determine all n such that Jack can make all coins show T after a finite number of operations.

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*Time: 4 hours 30 minutes
Each problem is worth 7 points*



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Day 2

Wednesday, February 17, 2021

Problem 1. Let n be a positive integer. Find the number of permutations $a_1, a_2, \dots, a_n$ of the sequence $1, 2, \dots, n$ satisfying

$$a_1 \leq 2a_2 \leq 3a_3 \leq \dots \leq na_n.$$

Problem 2. Find all positive integers n such that 2021^n can be expressed in the form $x^4 - 4y^4$ for some integers x, y .

Problem 3. A triangle ABC with $AB < AC < BC$ is given. The point P is the center of an excircle touching the line segment AB at D . The point Q is the center of an excircle touching the line segment AC at E . The circumcircle of the triangle ADE intersects $\overline{PE}$ and $\overline{QD}$ again at G and H respectively. The line perpendicular to $\overline{AG}$ at G intersects the side AB at R . The line perpendicular to $\overline{AH}$ at H intersects the side AC at S .

Prove that $\overline{DE}$ and $\overline{RS}$ are parallel.

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Day 3

Thursday, February 18, 2021

Problem 1. For each positive integer n , let $\rho(n)$ be the number of the positive divisors of n with exactly the same set of prime divisors as n .

Show that, for any positive integer m , there exists a positive integer n such that $\rho(202^n + 1) \geq m$.

Problem 2. Let $d \geq 1$ and $n \geq 0$ be integers. Find the number of ways to write down a nonnegative integer in each square of a $d \times d$ grid such that the numbers in any set of d squares, no two in the same row or column, sum to n .

Problem 3. Let m, n be positive integers. Show that the polynomial

$$f(x) = x^m(x^2 - 100)^n - 11$$

cannot be expressed as a product of two non-constant polynomials with integral coefficients.

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Day 4

Friday, February 19, 2021

Problem 1. Let a, b, c be distinct positive real numbers such that $\frac{1}{1+a} + \frac{1}{1+b} + \frac{1}{1+c} \leq 1$.
Prove that

$$2 \left(\sqrt{\frac{a+b}{ac}} + \sqrt{\frac{b+c}{ba}} + \sqrt{\frac{c+a}{cb}} \right) < \frac{a^3}{(a-b)(a-c)} + \frac{b^3}{(b-c)(b-a)} + \frac{c^3}{(c-a)(c-b)}.$$

Problem 2. Let n be a positive integer and let $0 \leq k \leq n$ be an integer. Show that there exist n points in the plane with no three on a line such that the points can be divided into two groups satisfying the following properties.

- (i) The first group has k points and the distance between any two distinct points in this group is irrational.
- (ii) The second group has $n-k$ points and the distance between any two distinct points in this group is an integer.
- (iii) The distance between a point in the first group and a point in the second group is irrational.

Problem 3. A finite sequence of integers $a_0, a_1, \dots, a_n$ is called *quadratic* if for each $i \in \{1, 2, \dots, n\}$ we have the equality $|a_i - a_{i-1}| = i^2$.

- (i) Prove that for any two integers b and c , there exist a positive integer n and a quadratic sequence with $a_0 = b$ and $a_n = c$.
- (ii) Find the smallest positive integer n for which there exists a quadratic sequence with $a_0 = 0$ and $a_n = 2021$.

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*Time: 4 hours 30 minutes
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